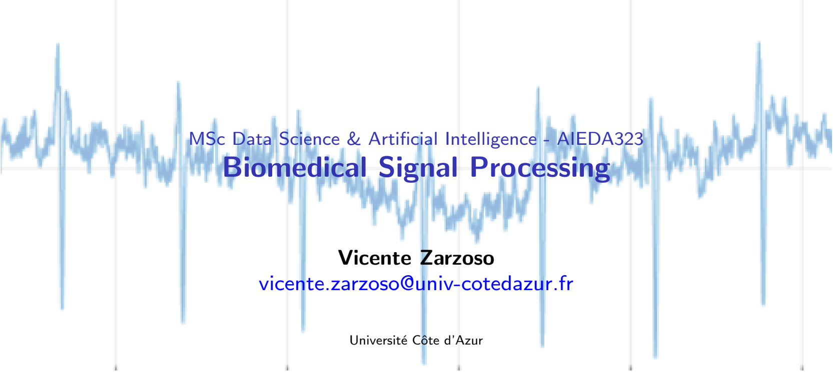

MSc Data Science & Artificial Intelligence - AIEDA323

Biomedical Signal Processing

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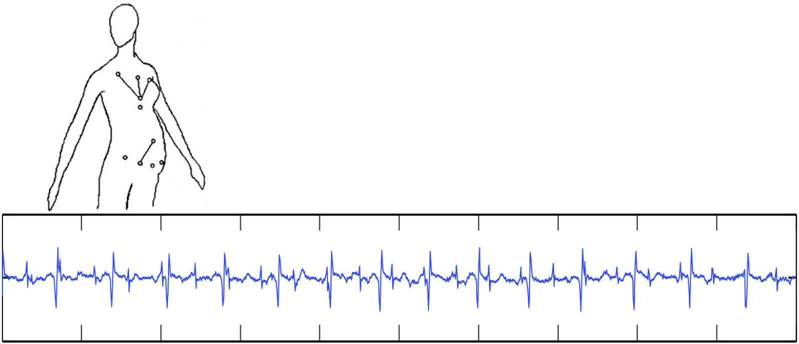
Chapter 3: Blind source separation

Objectives

- Acquire the **fundamentals of BSS** based on statistical independence
- Understand the basic **model assumptions and limitations**
- Understand **why operating blindly** is useful
- Become familiar with the **mathematical tools** for BSS
  - information-theoretic criteria, higher-order statistics (cumulants)
  - principal component analysis (PCA), **independent component analysis (ICA)**
- Study optimization methods giving rise to existing **practical algorithms** for BSS
  - PCA, ICA-CoM2, RobustICA
- Understand their **benefits and drawbacks**
- Illustrate their performance in **biomedical signal processing problems** [lab assignment]

Motivating example 1: Fetal ECG extraction

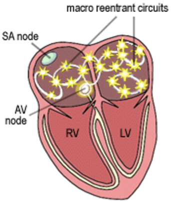
- Fetal ECG (FECG) is corrupted by maternal ECG (MECG) in surface records during pregnancy

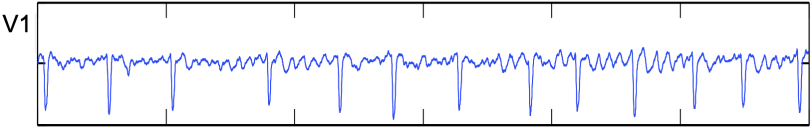


- How to process the available signals optimally to cancel the MECG and estimate the FECG?

Motivating example 2: Atrial activity extraction

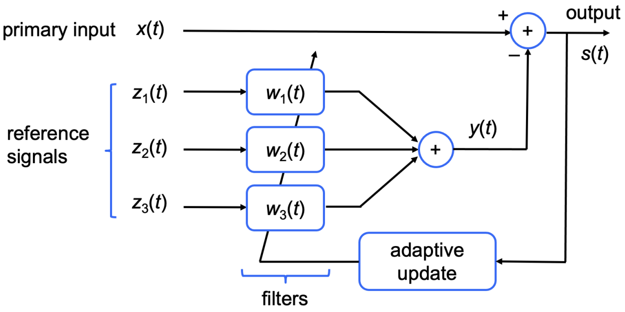
- Atrial activity (AA) is corrupted by ventricular activity (VA) in atrial fibrillation (AF) ECGs





- How to process the available signals optimally to cancel the VA and estimate the AA?

ANC — modeling assumptions



- Primary input: $x(t) = d(t) + b(t)$
  - $d(t)$: signal of interest (SOI), desired signal
  - $b(t)$: interference, decorrelated with the SOI
- Reference signals: $\{z_r(t)\}_{r=1}^R$
  - decorrelated with the SOI: $\mathrm{E}\{z_r(t)d(t)\} = 0$
  - correlated with the interference: $\mathrm{E}\{z_r(t)b(t)\} \neq 0$
- Estimated interference:

$$y(t) = \sum_{r=1}^{R} w_r(t) * z_r(t)$$

- Output signal:

$$s(t) = x(t) - y(t)$$

Modeling assumptions (cont'd)

Problem statement

Given the observed primary input $x(t)$ and reference signals $\{z_r(t)\}_{r=1}^R$, find filter impulse reponses $\{w_r(t)\}_{r=1}^R$ minimizing

$$\xi = \mathrm{E}\{s(t)^2\}$$

- Equivalent to minimizing $\mathrm{E}\{[s(t) - d(t)]^2\}$.

Key assumptions

A1) Desired signal $d(t)$ and interference $b(t)$ are uncorrelated:

$$\mathrm{E}\{d(t)b(t)\} = 0$$

A2) Reference signals $\{z_r(t)\}_{r=1}^R$ are uncorrelated with the desired signal:

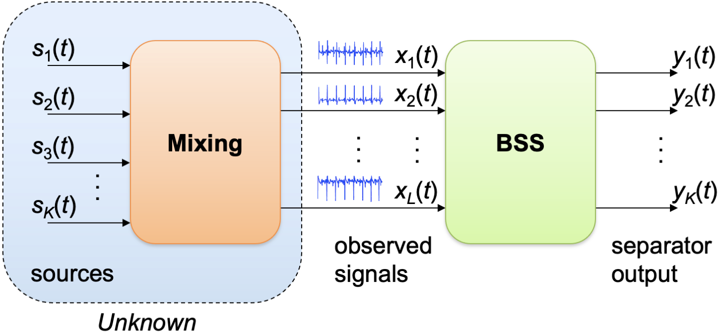
$$\mathrm{E}\{z_r(t)d(t)\} = 0, \quad r = 1, 2, \ldots, R \quad \Rightarrow \quad \mathrm{E}\{y(t)d(t)\} = 0$$

A3) At least one reference signal $z_r(t)$ is correlated with the interference:

$$\exists r \; : \; \mathrm{E}\{z_r(t)b(t)\} \neq 0$$

Q : What happens if assumption A2 is violated? **A**: Complete lab 2 to find out!

Blind source separation (BSS) model



Why blind?

- **Blind** means that the required prior knowledge about the sources and the mixture (channel) is kept to a minimum
  - no training sequences (communications) $\rightarrow$ improved spectral efficiency
  - no assumptions on model geometry (array manifold) $\rightarrow$ improved robustness to calibration errors
- By reducing model assumptions, blind processing is more robust than supervised approaches
  - improved robustness to modeling errors
  - increased versatility

Main advantage over MRANC approach

No need to optimize sensor placement to obtain reference signals containing contributions from interfering sources only.

The contribution of each source to each observation can be arbitrary as long as the mixing operation remains invertible.

Instantaneous linear mixtures

Each observation is an unknown linear combination of the source signals:

$$x_\ell(t) = \sum_{k=1}^{K} h_{\ell k} s_k(t), \quad 1 \le \ell \le L$$

It models propagation media without time dispersion.

Instantaneous linear mixing BSS model

$$\mathbf{x}(t) = \mathbf{H}\mathbf{s}(t) \tag{1}$$

$$\underbrace{\mathbf{x}(t) \stackrel{\text{def}}{=} [x_1(t), x_2(t), \ldots, x_L(t)]^{\mathrm{T}}}_{\textbf{observed signals}} \qquad \underbrace{\mathbf{s}(t) \stackrel{\text{def}}{=} [s_1(t), s_2(t), \ldots, s_K(t)]^{\mathrm{T}}}_{\textbf{source signals}} \qquad \underbrace{[\mathbf{H}]_{\ell k} \stackrel{\text{def}}{=} h_{\ell k}}_{\textbf{mixing matrix}}$$

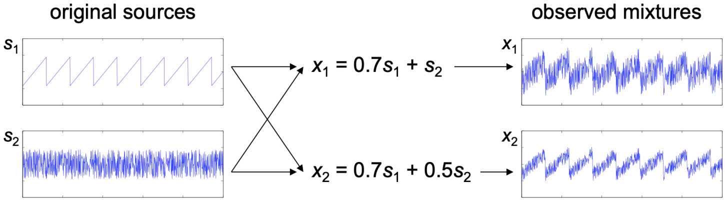
Goals

From N time samples of the observed signals $\mathbf{x}(t)$, $t = 0, 1, \ldots, N-1$:

- **Source recovery**: estimate the source signals $\mathbf{s}(t)$, $t = 0, 1, \ldots, N-1$.
- **Mixture (or channel) identification**: estimate the mixing matrix $\mathbf{H}$.

Instantaneous linear mixtures (cont'd)

Example 1: Synthetic (2×2) mixture



BSS: recover $s_1(t)$ and $s_2(t)$ from the only knowledge of $x_1(t)$ and $x_2(t)$, with no knowledge on the mixing coefficients.

Instantaneous linear mixture model (cont'd)

- **Overdetermined mixtures**
  - at least as many sensors as sources: $L \geq K$
  - if $\mathbf{H}$ is full column rank, it's invertible
  - applying $\mathbf{H}^{-1}$ (the left pseudo-inverse of $\mathbf{H}$) to the observations recovers the sources

    $\rightarrow$ problems of source recovery and channel identification are equivalent
  - **Inverse filter approach**

    Estimate a separating matrix $\mathbf{W}$ such that $\mathbf{y}(t) = \mathbf{W}^{\mathsf{T}}\mathbf{x}(t) = \mathbf{s}(t) \quad \rightarrow \quad \mathbf{W}^{\mathsf{T}} = \mathbf{H}^{-1}$

- **Underdetermined mixtures**
  - fewer sensors than sources: $L < K$
  - matrix $\mathbf{H}$ is no longer left-invertible
  - even if channel is identified, source recovery is not unique.

Identifiability

For any arbitrary invertible matrix $\mathbf{A} \in \mathbb{R}^{K \times K}$, define:

$$\tilde{\mathbf{H}} = \mathbf{H}\mathbf{A} \qquad \tilde{\mathbf{s}}(t) = \mathbf{A}^{-1}\mathbf{s}(t)$$

Then:

$$\tilde{\mathbf{H}}\tilde{\mathbf{s}}(t) = (\mathbf{H}\mathbf{A})(\mathbf{A}^{-1}\mathbf{s}(t)) = \mathbf{H}\underbrace{(\mathbf{A}\mathbf{A}^{-1})}_{\mathbf{I}_K}\mathbf{s}(t) = \mathbf{H}\mathbf{s}(t) = \mathbf{x}(t)$$

Couples $\big(\mathbf{H}, \mathbf{s}(t)\big)$ and $\big(\tilde{\mathbf{H}}, \tilde{\mathbf{s}}(t)\big)$ yield exactly the same observations.

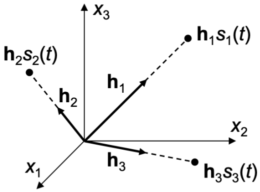
BSS is an ill-posed problem.

Further assumptions are necessary on $\mathbf{H}$ and/or $\mathbf{s}(t)$ to make the separation possible.

Basic assumptions for source recovery

$$\mathbf{x}(t) = \mathbf{H}\mathbf{s}(t) = \sum_{k=1}^{K} \mathbf{h}_k s_k(t)$$

$\mathbf{h}_k$: kth column of the mixing matrix
(**source direction**, **spatial factor**)



Example 2: Suppose that two sources, say s_1 and s_2, share the same mixing column, $\mathbf{h}_2 = a\mathbf{h}_1$.

$$\mathbf{x}(t) = \sum_{k=1}^{K} \mathbf{h}_k s_k(t) = \mathbf{h}_1[\underbrace{s_1(t) + as_2(t)}_{\tilde{s}_1(t)}] + \sum_{k=3}^{K} \mathbf{h}_k s_k(t) = \mathbf{h}_1 \tilde{s}_1(t) + \sum_{k=3}^{K} \mathbf{h}_k s_k(t)$$

Sources s_1 and s_2 contribute to the mixture along the same direction $\mathbf{h}_1$. The model 'sees' a source mixture $\tilde{s}_1 = s_1 + as_2$ rather than two different sources. As a result, the two sources cannot be distinguished.

Necessary condition for source recovery

Mixing matrix $\mathbf{H}$ must have linearly independent columns, i.e., it must be full column rank, which implies $L \geq K$ (overdetermined mixtures).

Essential identifiability or uniqueness

- **Scale ambiguity**: a scale factor may be exchanged between a source signal $s_k(t)$ and its direction $\mathbf{h}_k$ without altering the observations

$$\mathbf{h}_k s_k(t) = (\mathbf{h}_k \alpha)(s_k(t)/\alpha), \qquad \alpha \neq 0$$

- **Permutation ambiguity**: source ordering is immaterial

$$\mathbf{x}(t) = [\mathbf{h}_1 \ \mathbf{h}_2][s_1(t) \ s_2(t)]^\mathsf{T} = \mathbf{h}_1 s_1(t) + \mathbf{h}_2 s_2(t) = \mathbf{h}_2 s_2(t) + \mathbf{h}_1 s_1(t) = [\mathbf{h}_2 \ \mathbf{h}_1][s_2(t) \ s_1(t)]^\mathsf{T}$$

Essential uniqueness

Separating matrix is allowed to fulfill $\mathbf{W}^\mathsf{T}\mathbf{H} = \mathbf{PD}$

$\mathbf{P}$: permutation matrix $\qquad$ $\mathbf{D}$: diagonal matrix

$\rightarrow$ estimated sources: scaled, permuted versions of the original sources.

To reduce the scale ambiguity, the sources can be assumed to have unit variance:

$$\mathrm{E}\{s_k^2\} = 1, \quad k = 1, 2, \ldots, K \tag{2}$$

In that case, $[\mathbf{D}]_{kk} = \pm 1$.

Instantaneous linear mixture model (cont'd)

Basic separation principle: property retrieval

- Recover the sources by recovering some measurable property of the sources
- These are typically *"non properties"*:
  - Non-dependence, non-Gaussianity → PCA, ICA
  - Non-whiteness or non-i.i.d. (time coherence, spectral color)
  - Non-stationarity
  - Non-negativity
- But others are also exploitable:
  - sparsity
  - periodicity, harmonicity
  - known discrete support, finite alphabet (digital communications)
  - known dominant support, positive support
  - ...

Approach: find a separating matrix $\mathbf{W}$ such the separator output $\mathbf{y}(t) = \mathbf{W}^{\mathsf{T}}\mathbf{x}(t)$ inherits the source properties.

Under suitable *identifiability conditions*, recovering the assumed source property at the separator output leads to the recovery of the sources themselves.

Why statistical independence?

- Reasonable assumption in many applications
- Signals arising from physically independent phenomena are also statistically independent

Examples

- communications
  - transmitted signals from different users in wireless systems
  - ...
- biomedical
  - maternal and fetal heartbeats during pregnancy
  - atrial and ventricular activities in atrial fibrillation
  - brain region activations in magnetic resonance imaging
  - ...

Exploiting second-order statistics: PCA

Idea: recover the source second-order statistical structure.

Under the statistical independence assumption, the sources are in particular second-order independent (uncorrelated):

$$\boxed{\mathrm{E}\{s_i s_j\} = 0, \quad i \neq j} \tag{3}$$

Combining eqns (2)–(3):

$$\boxed{\mathbf{R}_s \stackrel{\text{def}}{=} \mathrm{E}\{\mathbf{s}\mathbf{s}^\mathsf{T}\} = \mathbf{I}_K} \tag{4}$$

$\rightarrow$ Find a separating matrix $\mathbf{W}$ such the separator output $\mathbf{y}(t) = \mathbf{W}^\mathsf{T}\mathbf{x}(t)$ has an identity covariance matrix:

$$\boxed{\mathbf{R}_y = \mathbf{R}_s = \mathbf{I}_K}$$

Exploiting second-order statistics: PCA

From the BSS model:

$$\mathbf{R}_x \stackrel{\text{def}}{=} \mathrm{E}\{\mathbf{x}\mathbf{x}^\mathsf{T}\} \underset{\substack{\uparrow \\ \text{eqn. (1)}}}{=} \mathrm{E}\{\mathbf{H}\mathbf{s}\mathbf{s}^\mathsf{T}\mathbf{H}^\mathsf{T}\} = \mathbf{H}\mathrm{E}\{\mathbf{s}\mathbf{s}^\mathsf{T}\}\mathbf{H}^\mathsf{T} \underset{\substack{\uparrow \\ \text{eqn. (4)}}}{=} \mathbf{H}\mathbf{H}^\mathsf{T}$$

Eigenvalue decomposition (EVD):

$$\mathbf{R}_x = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\mathsf{T}$$

Comparing the above expressions, we can identify:

PCA solution to the BSS problem

$$\boxed{\hat{\mathbf{H}} = \mathbf{U}\boldsymbol{\Lambda}^{1/2}} \qquad \boxed{\hat{\mathbf{W}} \stackrel{\text{def}}{=} \hat{\mathbf{H}}^{-\mathsf{T}} = \mathbf{U}\boldsymbol{\Lambda}^{-1/2}}$$

$$\mathbf{y}(t) = \hat{\mathbf{W}}^\mathsf{T}\mathbf{x}(t) = \boldsymbol{\Lambda}^{-1/2}\mathbf{U}^\mathsf{T}\mathbf{x}(t)$$

The source estimates correspond to the **normalized principal components** of the observations.

→ *Exercise 1:* Check that source uncorrelation is restored, i.e., $\mathbf{R}_y = \mathbf{R}_s = \mathbf{I}_K$.

Computing PCA from the SVD of the observation matrix

$$\mathbf{X} \stackrel{\text{def}}{=} [\mathbf{x}(0), \mathbf{x}(1), \ldots, \mathbf{x}(N-1)] \in \mathbb{R}^{K \times N} \underset{\substack{\uparrow \\ \text{SVD}}}{=} \mathbf{U}_x \mathbf{\Lambda}_x \mathbf{V}_x^{\mathsf{T}}$$

$$\mathbf{R}_x \approx \frac{1}{N}\mathbf{X}\mathbf{X}^{\mathsf{T}} = \frac{1}{N}\mathbf{U}_x \mathbf{\Lambda}_x^2 \mathbf{U}_x^{\mathsf{T}}$$

$$\mathbf{R}_x = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^{\mathsf{T}} = \mathbf{H}\mathbf{H}^{\mathsf{T}}$$

$$\mathbf{U} \approx \mathbf{U}_x \qquad \mathbf{\Lambda} \approx \frac{1}{N}\mathbf{\Lambda}_x^2$$

PCA solution from observation matrix

$$\hat{\mathbf{H}} = \frac{1}{\sqrt{N}}\mathbf{U}_x \mathbf{\Lambda}_x \qquad \hat{\mathbf{W}} = \hat{\mathbf{H}}^{-1} = \sqrt{N}\,\mathbf{\Lambda}_x^{-1}\mathbf{U}_x^{\mathsf{T}}$$

$$\mathbf{Y} = \hat{\mathbf{W}}\mathbf{X} = \sqrt{N}\,\mathbf{V}_x^{\mathsf{T}}$$

$$\mathbf{Y} \stackrel{\text{def}}{=} [\mathbf{y}(0), \mathbf{y}(1), \ldots, \mathbf{y}(N-1)] \in \mathbb{R}^{K \times N}$$

→ *Exercise 2:* Check that source uncorrelation is restored, i.e., $\mathbf{R}_y \approx \frac{1}{N}\mathbf{Y}\mathbf{Y}^{\mathsf{T}} = \mathbf{I}_K$.

Maximum variance formulation: deflation PCA

In an alternative yet equivalent formulation, PCA can be cast as the search for successive projections with maximal variance.

PCA: maximum variance formulation

- Find $\tilde{\mathbf{w}}_1 \in \mathbb{R}^K$ maximizing $\mathrm{E}\{\tilde{y}_1^2\}$, with $\tilde{y}_1 = \tilde{\mathbf{w}}_1^\mathsf{T}\mathbf{x}$, subject to $\|\tilde{\mathbf{w}}_1\| = 1$
- Find $\tilde{\mathbf{w}}_2 \in \mathbb{R}^K$ maximizing $\mathrm{E}\{\tilde{y}_2^2\}$, with $\tilde{y}_2 = \tilde{\mathbf{w}}_2^\mathsf{T}\mathbf{x}$, subject to $\|\tilde{\mathbf{w}}_2\| = 1$, $\tilde{\mathbf{w}}_2^\mathsf{T}\tilde{\mathbf{w}}_1 = 0$

  $\vdots$
- Find $\tilde{\mathbf{w}}_K \in \mathbb{R}^K$ maximizing $\mathrm{E}\{\tilde{y}_K^2\}$, with $\tilde{y}_K = \tilde{\mathbf{w}}_K^\mathsf{T}\mathbf{x}$, subject to $\|\tilde{\mathbf{w}}_K\| = 1$, $\tilde{\mathbf{w}}_K^\mathsf{T}\tilde{\mathbf{w}}_j = 0$, $j < K$

$\{\tilde{\mathbf{w}}_k\}_{k=1}^K$: **principal directions** $\{\tilde{y}_k\}_{k=1}^K$: (unnormalized) **principal components**

Up to irrelevant scale factors, results are identical to the EVD of $\mathbf{R}_x$:

$$\tilde{\mathbf{W}} = \mathbf{\Lambda}^{1/2}\hat{\mathbf{W}} = \mathbf{U} \qquad \tilde{\mathbf{y}} = \mathbf{\Lambda}^{1/2}\mathbf{y} = \mathbf{U}^\mathsf{T}\mathbf{x}$$

but principal components are computed one after another. This approach is called **deflation**.

Why second-order statistics are not enough?

→ *Exercise 3:* Prove that, for any unitary matrix $\mathbf{Q} \in \mathbb{R}^{K\times K}$, $\mathbf{R}_{\hat{s}} = \mathbf{I}_K$, where $\hat{\mathbf{s}}(t) = \mathbf{Qs}(t)$.

This means that, in general, recovering the identity covariance matrix structure at the separator output will leave an indeterminacy in the form of a unitary mixing matrix $\mathbf{Q}$ that cannot be identified without exploiting further information.

Whitening or sphering

The covariance matrix diagonalization performed by PCA is also called **whitening** or **sphering**.

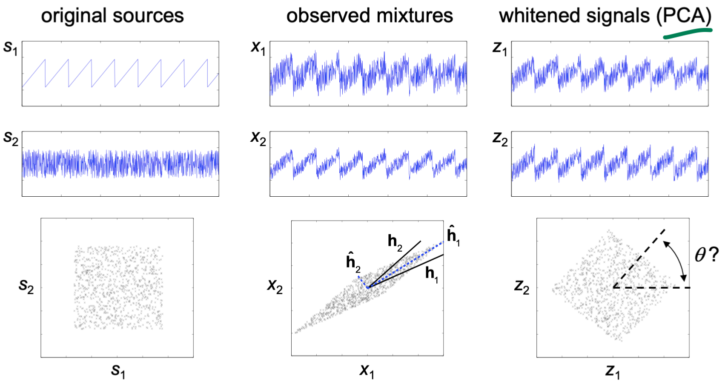
It reduces the mixing to an unknown unitary matrix $\mathbf{Q}$:

$$\boxed{\mathbf{z}(t) \stackrel{\text{def}}{=} \mathbf{y}_{\text{PCA}}(t) = \mathbf{Qs}(t)}$$

→ *Exercise 4:* In which cases will PCA succeed to perform source separation? Which constraints need to be respected in the observation model? Are they plausible in a biomedical context?

Why second-order statistics are not enough? (cont'd)

Example 3: Synthetic (2×2) mixture — continued from [Example 1]



Scatter plot: points $\big(s_1(t), s_2(t)\big)$, $t = 0, 1, \ldots, N-1 \rightarrow$ sample approximation of the probability density function (pdf) $p_s(s_1, s_2)$.

Exploiting higher-order statistics: ICA

Idea: recover the source higher-order statistical structure.

Independent component analysis (ICA)

Find a linear transformation **W** maximizing the statistical independence between the separator output components.

The concept of ICA was coined by Pierre Comon in the 1990s.[1]

[1] P. Comon, "Independent component analysis, a new concept?", *Signal Processing*, 36(3):287–314, 1994.

Measuring statistical independence

Kullback-Leibler divergence

$$KL(p_x \mid p_y) \stackrel{\text{def}}{=} \int_{\mathcal{S}} p_x(\mathbf{u}) \log \frac{p_x(\mathbf{u})}{p_y(\mathbf{u})} d\mathbf{u}$$

$$KL(p_x \mid p_y) \geq 0, \qquad KL(p_x \mid p_y) = 0 \quad \Leftrightarrow \quad p_x = p_y$$

Mutual information

$$I(\mathbf{y}) \stackrel{\text{def}}{=} KL\left(p_y \,\middle|\, \prod_{k=1}^{K} p_{y_k}\right) = \int_{\mathcal{Y}} p_y(\mathbf{u}) \log \frac{p_y(\mathbf{u})}{\prod_{k=1}^{K} p_{y_k}(u_k)} d\mathbf{u}$$

$I(\mathbf{y}) \geq 0, \qquad I(\mathbf{y}) = 0 \quad \Leftrightarrow \quad \mathbf{y}$ is made of independent components:

$$p_y(\mathbf{u}) = \prod_{k=1}^{K} p_{y_k}(u_k)$$

$$I(\mathbf{y}) = \sum_{k=1}^{K} H(y_k) - H(\mathbf{y}) \quad \text{with} \begin{cases} H(\mathbf{y}) = -\int_{\mathcal{Y}} p_y(\mathbf{u}) \log p_y(\mathbf{u}) d\mathbf{u} : & \textbf{Shanon entropy} \\ H(y_k) = -\int_{\mathcal{Y}_k} p_{y_k}(u) \log p_{y_k}(u) du : & \textbf{marginal entropy} \end{cases}$$

Measuring statistical independence after whitening

$$I(\mathbf{y}) = \sum_{k=1}^{K} H(y_k) - H(\mathbf{y})$$

$$\mathbf{z}(t) = \mathbf{Q}\mathbf{s}(t), \quad \mathbf{Q} \text{ unitary}$$

Property: joint entropy $H(\mathbf{y})$ is not modified by unitary transformations

$$\min I(\mathbf{y}) \quad \Leftrightarrow \quad \min E(\mathbf{y})$$

$$E(\mathbf{y}) = \sum_{k=1}^{K} H(y_k) = -\sum_{k=1}^{K} KL(p_{y_k} \mid g), \quad g \sim \mathcal{N}(0,1)$$

Independence criteria under whitening assumption

Minimizing mutual information is equivalent to **minimizing the sum of marginal entropies** of the separator output components.

In turn, this is equivalent to **maximizing** the **non-Gaussianity** of the components.

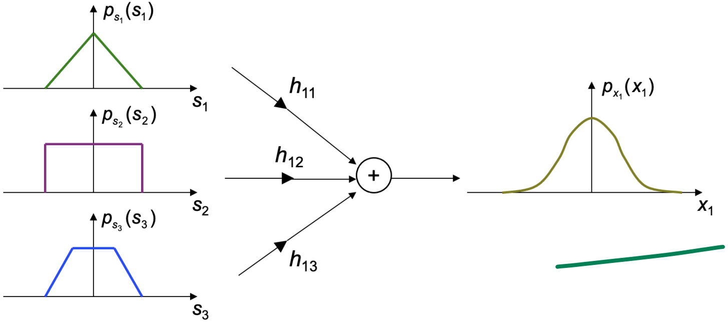
Why non-Gaussianity?

Central limit theorem (CLT)

Let $\{s_k\}_{k=1}^{K}$ be zero-mean, unit-variance, independent random variables. Then:

$$\frac{1}{\sqrt{K}} \sum_{k=1}^{K} s_k \xrightarrow[K\to\infty]{\mathrm{d}} \mathcal{N}(0,1)$$

Nontrivial mixing increases Gaussianity $\rightarrow$ marginal entropy increases under unitary mixing.



Practical ICA algorithms based on cumulants

Cumulants[2]

- Coefficients of the Taylor expansion of the second characteristic function:

$$\mathrm{cum}(x_{i_1}, x_{i_2}, \ldots x_{i_r}) \stackrel{\mathrm{def}}{=} (-j)^r \left. \frac{\partial^r \log \mathrm{E}\{\exp(j\mathbf{u}^\mathsf{T}\mathbf{x})\}}{\partial u_{i_1} \partial u_{i_2} \ldots \partial u_{i_r}} \right|_{\mathbf{u}=\mathbf{0}}$$

- If $r > 2$: *higher-order cumulants*
- *Cross-cumulants* involve more than one variable: $\exists k \neq \ell \;:\; i_k \neq i_\ell$
- *Marginal cumulants* involve a single variable: $i_k = i$, $\forall k = 1, 2, \ldots, r$
- For *zero-mean* random variables:

$$\begin{aligned}
\mathrm{cum}(x_i, x_j) &= \mathrm{E}\{x_i x_j\} \text{ : covariance of } x_i \text{ and } x_j \\
\mathrm{cum}(x_i, x_j, x_k, x_\ell) &= \mathrm{E}\{x_i x_j x_k x_\ell\} - \mathrm{E}\{x_i x_j\}\mathrm{E}\{x_k x_\ell\} - \mathrm{E}\{x_i x_k\}\mathrm{E}\{x_j x_\ell\} - \mathrm{E}\{x_i x_\ell\}\mathrm{E}\{x_j x_k\} \\
\mathrm{cum}(x, x) &= \mathrm{E}\{x^2\} = \sigma_x^2 \text{ : variance of } x \\
\mathrm{cum}(x, x, x, x) &= \mathrm{E}\{x^4\} - 3\mathrm{E}\{x^2\}^2 \stackrel{\mathrm{def}}{=} \kappa_x \text{ : fourth-order cumulant of } x
\end{aligned}$$

[2] A. Stuart, K. Ord, *Kendall's Advanced Theory of Statistics*, 6th Ed., Hodder Arnold, 1994.

Practical ICA algorithms based on cumulants (cont'd)

Cumulants and statistical independence

The entries of a random vector $\mathbf{x}$ are independent up to order r if and only if all their cross-cumulants of order $n \leq r$ are null.

Cumulants and Gaussianity

The higher-order cumulants ($r > 2$) of a Gaussian random variable are null.

Gram-Charlier expansion

A pdf can be approximated using cumulants.

For a symmetric zero-mean random variable:

$$p_s(s) \approx g(s) \left\{ 1 + \frac{(\sigma_s^2 - 1)}{2!} h_2(s) + \frac{\kappa_s}{4!} h_4(s) \right\} \qquad g(s) = \frac{1}{\sqrt{2\pi}} \exp(-s^2/2)$$

$h_n(s)$: Hermite polynomial of degree n

$$h_n(s) \stackrel{\text{def}}{=} \frac{(-1)^n}{g(s)} \frac{d^n g(s)}{ds^n} \quad \text{e.g.,} \begin{cases} h_2(s) = s^2 - 1 \\ h_4(s) = s^4 - 6s^2 + 3 \end{cases}$$

Practical ICA algorithms based on cumulants (cont'd)

Using pdf approximations under the whitening constraint:

$$I(\mathbf{y}) = E(\mathbf{y}) \approx -\sum_{k=1}^{K} \kappa_{y_k}^2$$

$$\min I(\mathbf{y}) \quad \Leftrightarrow \quad \min E(\mathbf{y}) \quad \Leftrightarrow \quad \text{maximize the sum of squared marginal cumulants}$$

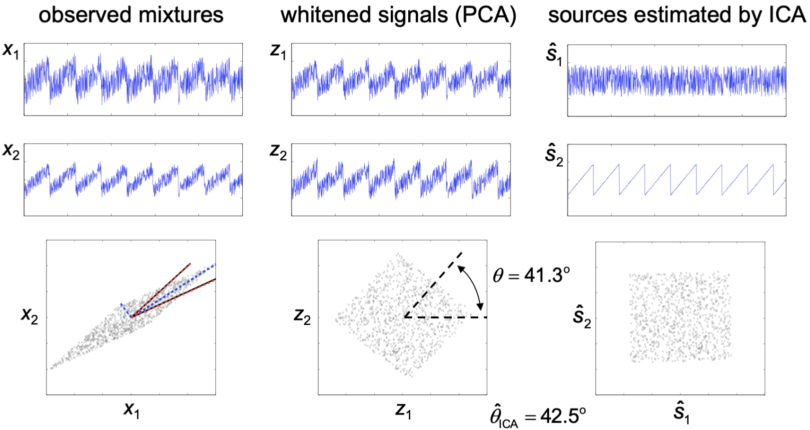
Comon's ICA (CoM2): Jacobi-like pairwise mutual information mimimization

1. Initialize $\mathbf{y} = \mathbf{z}$ (whitened signals)
2. For each index pair (i, j), $i < j$:
3. Apply planar Givens rotation $\mathbf{Q}(\theta_{ij})^{\mathsf{T}}$ to signal pair $\mathbf{y}_{ij} = [y_i, y_j]^{\mathsf{T}}$
   - angle θ_{ij} maximizes $\Psi(\mathbf{Q}(\theta_{ij})^{\mathsf{T}}\mathbf{y}_{ij}) = \kappa_{y_i'}^2 + \kappa_{y_j'}^2$
   - roots of a quartic (4th-degree polynomial)
   - update $\mathbf{y}_{ij} \leftarrow \mathbf{Q}(\theta_{ij})^{\mathsf{T}}\mathbf{y}_{ij}$
4. Sweep over all signal pairs until convergence

Complexity: $O(K^{5/2}N)$

Practical ICA algorithms based on cumulants (cont'd)

Example 4: Synthetic (2×2) mixture — continued from [Example 1] and [Example 3]



Deflationary separation

Joint separation

Sources are extracted simultaneously by estimating the entire separating matrix $\mathbf{W}$:

$$\hat{\mathbf{s}}(t) = \mathbf{y}(t) = \mathbf{W}^{\mathsf{T}}\mathbf{x}(t)$$

Deflationary separation

Sources are extracted iteratively one after another by

- estimating a column vector $\mathbf{w}_k$ of $\mathbf{W}$ (extracting vector) at a time

$$\hat{s}_k(t) = y_k(t) = \mathbf{w}_k^{\mathsf{T}}\mathbf{x}(\mathbf{t})$$

- subtracting the estimated source contribution from the observations

  or

- assuring that the next extracting vector is linearly independent from the previously obtained extracting vectors (e.g., via Gram-Schmidt orthogonalization)

Deflationary separation (cont'd)

Kurtosis maximization (KM) criterion

$$\psi_{\mathrm{KM}}(y) = \frac{|\kappa_y|}{(\sigma_y^2)^2}$$

$$\kappa_y = \mathrm{E}\{|y|^2\} - 2\mathrm{E}\{|y|\}^2 - |\mathrm{E}\{y^2\}|^2$$

Advantages[3]

- Lack of spurious extrema in ideal conditions $\rightarrow$ guaranteed convergence to an independent component
- Valid for real-valued as well as complex-valued sources, even with non-circular distributions ($\mathrm{E}\{y^2\} \neq 0$)
- Whitening is not required $\rightarrow$ improved asymptotic performance
- Can target sub-Gaussian ($\kappa_s < 0$) or super-Gaussian sources ($\kappa_s > 0$)

[3] V. Zarzoso, P. Comon, "Robust independent component analysis by iterative maximization of the kurtosis contrast with algebraic optimal step size", *IEEE Transactions on Neural Networks*, 21(2):248–261, 2010.

Deflationary separation (cont'd)

RobustICA: exploit the structure of the KM criterion

- Use original kurtosis contrast ψ_{KM} without simplifications
- At each iteration i, perform exact one-dimensional search:

$$\mathbf{w}^{(i+1)} = \mathbf{w}^{(i)} + \mu \mathbf{g}^{(i)} \qquad \mathbf{g}^{(i)} \stackrel{\text{def}}{=} \nabla \psi_{\mathsf{KM}}(\mathbf{w}^{(i)})$$

$$\mu_{\text{opt}}^{(i)} = \arg\max_{\mu} \psi_{\mathsf{KM}}(\mathbf{w}^{(i)} + \mu \mathbf{g}^{(i)})$$

- Optimal step-size candidates: solutions of

$$\frac{\partial \psi_{\mathsf{KM}}(\mathbf{w}^{(i)} + \mu \mathbf{g}^{(i)})}{\partial \mu} \stackrel{\text{def}}{=} p^{(i)}(\mu) = 0$$

KM criterion: $p^{(i)}(\mu)$ is a 4th-degree polynomial in μ $\quad\rightarrow\quad$ μ_{opt} can be determined *algebraically* at each iteration.

Deflationary separation (cont'd)

Benefits

- improved robustness to saddle points and spurious local extrema (short sample size, noise)
- excellent source extraction quality vs. number of operations tradeoff
- idea easily adapted to other blind and semi-blind criteria

Drawbacks

- convergence speed decreases as data dimension increases

Freely available MATLAB implementation:

https://i3s.univ-cotedazur.fr/~zarzoso/robustica.html

Algorithm implementation

→ *Exercise 5:*

1. Create a function `bss_pca.m` implementing the PCA algorithm for BSS (whitening)
   - **Input parameters**

observed signals	matrix `X` with dimensions $K \times N$, one signal per row

   - **Output parameters**

estimated sources	matrix `S` with dimensions $K \times N$, one signal per row
estimated mixing matrix	matrix `H` with dimensions $K \times K$

2. Generate a synthetic signal set:
   - Retrieve the signals generated in [Chapter 2 ANC, Exercise 5].
     Consider the primary signal $x(t)$ and the reference signals $z_1(t)$ and $z_2(t)$ as the output of 3 sensors.

3. Using function `bss_pca.m`, obtain the PCA sources (whitened signals). Plot the obtained sources. Identify the source of interest and compare it visually with the desired signal.

Algorithm implementation (cont'd)

→ *Exercise 6:* Evaluate the desired signal estimation quality in the results of the previous exercise:

1. Show the estimated mixing matrix. Check the orthogonality of its columns for PCA. What is the mixing matrix theoretically expected?
2. Compute the contribution of the desired signal to the primary signal (first sensor).
3. Plot and compare the desired signal, the observed signal in the first sensor and the estimation of the desired signal.
4. Compute and plot the error signal, defined as the difference between the desired signal and its estimate.
5. Compute, plot and compare the frequency spectrum (dB) of the desired signal, the observed signal and the estimated signal.
6. Compute and compare the power of the desired signal, the observed signal, the estimated signal and the error signal.
7. Estimate the signal-to-interference ratio (SIR) of the estimated signal and the SIR improvement brought about by BSS-PCA.

Algorithm implementation (cont'd)

→ *Exercise 7:*

1. Download and install the MATLAB package of the `RobustICA algorithm`.
   Using the `help` command, learn about the different functions contained in the package (call, input/output parameters, etc.)

2. Repeat [Exercise 5] and [Exercise 6] using RobustICA with prewhitening and deflationary orthogonalization:

```
deftype = 'o'     dimred = 0     Wini = []
kurtsign = []     tol = 1e-3     max_it = 1e3
```

Implementation tips

Estimating the source contributions to the observations

- BSS yields estimates of mixing matrix $\hat{\mathbf{H}}$ and source signals $\hat{\mathbf{S}}$ in model

$$\mathbf{X} = \mathbf{H}\mathbf{S}$$

  $\hat{\mathbf{H}}(i, j)$: coefficient with which jth source contributes to ith observation
  $\hat{\mathbf{S}}(i, j) = \hat{s}_i(j)$: ith estimated source signal at sample instant j

- To isolate the contribution of source k to all observations, compute:

$$\mathbf{X}_k = \hat{\mathbf{H}}(:, k)\hat{\mathbf{S}}(k, :)$$

  $\mathbf{X}_k(i, j)$: contribution of kth estimated source to jth time sample of ith observation, $x_i(j)$

$\rightarrow$ *Exercise 8:* How to compute the contribution of sources k_1 and k_2 to observation $x_n(t)$?

Summary

- BSS is an important signal processing problem, arising in a wide variety of applications
- Blind approach is more robust and flexible than supervised approach
  - improved robustness to sensor placement in biomedical applications
- Sources can be recovered by restoring one (or more) of their properties
  - independence / non-Gaussianity $\rightarrow$ ICA
- Statistical independence is an important BSS criterion
  - reasonable assumption in many practical problems
  - closely linked to solid information-theoretic principles:
    - mutual information and marginal entropy minimization
- Covariance diagonalization or whitening (PCA) is typically insufficient for BSS
  - but simplifies further processing
- Practical ICA algorithms can be developed by means of cumulant-based pdf expansions
  - Comon's CoM2
- Deflationary separation based on kurtosis maximization leads to globaly convergent algorithms in ideal conditions
  - RobustICA